Team Report on Data Collection for Kitwe-Focused News Project

Introduction
The data collection phase for the Kitwe-Focused News Project was designed to gather local news data from reputable Zambian news websites. Our primary objective was to create a dataset specifically related to Kitwe, Zambia, by targeting relevant news articles across multiple sources. This report details the data collection methodology, tools and processes used, challenges encountered, and limitations observed during the process.

Data Collection Approach
To collect Kitwe-centered news data, we adopted a systematic approach focused on accessing RSS (Really Simple Syndication) feeds from several Zambian news websites. RSS feeds are standardized XML formats that allow websites to share real-time updates. We applied this approach by configuring a Python script to extract data specifically tagged with “Kitwe” in the RSS feeds, ensuring that only relevant content was gathered. The script iterated over multiple pages from each source, capturing a broad range of articles. This strategy helped us secure a substantial dataset, including content from well-known sources like Lusaka Times, all filtered to prioritize Kitwe.

Tools and Methods
The data collection process utilized a range of tools and libraries, with Python serving as the primary programming language. The feed parser library was integral to the project, allowing us to extract content from the RSS feeds seamlessly. Additionally, the CSV library in Python was used to store collected data in a structured CSV file, allowing for easy data manipulation and later analysis. The CSV format enabled each article to be saved with detailed attributes, including the news source, category tags, headline, link, description, publication date, and author. This data structure allowed us to maintain organized, comprehensive records for each news entry.

The script was configured to retrieve up to 1,000 pages per news source, maximizing data collection from each outlet. This strategy was instrumental in capturing a wide range of articles, ensuring robust coverage of Kitwe-related news.

Challenges Encountered
During the data collection phase, several challenges arose. One key challenge was inconsistent data availability. Some RSS feeds did not consistently provide data for every field, leading to gaps in critical information like publication dates or descriptions. Additionally, pagination proved complex; some sources used irregular pagination, causing the script to prematurely stop retrieving data in certain cases.

Tagging and categorization presented further issues. Many articles lacked detailed tags or categories, complicating our attempts to classify news by topic. Connection issues also emerged periodically, with interruptions in accessing RSS feeds causing delays. Although retry mechanisms were implemented to handle these interruptions, not all sources provided stable connections.

Duplicate entries were another notable issue. Since RSS feeds occasionally repeated articles across pages, we needed to filter for duplicates, a step crucial to maintaining data integrity. These challenges highlighted areas for potential improvement in both the script and the data collection process.

Limitations

There were limitations that impacted the comprehensiveness of the dataset. One limitation was our reliance on RSS feeds; the data collection was restricted to articles that were available through these feeds. If a news item was not indexed by the RSS feed, it would not appear in our dataset. This reliance may have led to incomplete data coverage.

Another limitation was the occurrence of incomplete entries. Some articles lacked full descriptions or sufficient metadata, making deeper analysis more challenging. Additionally, since the scope was centered around Kitwe, there is an inherent source bias in the data. This project targeted Kitwe news specifically, meaning that broader regional news coverage was not included.

Finally, while automated processes handled most of the data retrieval, some manual quality checks were necessary. These checks, though essential, introduced the potential for human error.

Conclusion

The data collection phase of the Kitwe-Focused News Project was successful in creating a foundational dataset of local news articles, capturing insights into news trends in Kitwe. Despite facing challenges with data consistency, connectivity, and duplicate content, the systematic use of RSS feeds allowed for an efficient data collection process. The completed CSV file provides a reliable resource for the next phases, supporting a deeper understanding of local news dynamics centered around Kitwe. This data will now proceed to the preprocessing and analysis stages, where it will further support efforts to analyze and interpret local news patterns in the community.